Sage and CivicBench: Building and Evaluating Civic AI for Wildfire Recovery

CS 150: Generative AI for Social Impact

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1. Introduction

As large language models are increasingly used to give advice, applying them to civic contexts introduces a different set of challenges. Civic decision-making is not just about accessing information. It involves understanding how institutions work, identifying where power lies, and figuring out how individuals and groups can act effectively within those systems. In practice, this often means navigating complex processes, weighing different strategies, and making decisions that balance individual needs with broader community outcomes, especially in high-stakes situations like wildfire recovery.

In these contexts, the challenge is not simply answering a question, but helping people think through what they should do next. Existing resources provide general information, but they rarely support the kind of strategic reasoning, tradeoff evaluation, and collective action that civic engagement requires. At the same time, while many benchmarks exist for evaluating large

language models, there is no standard way to assess whether a system can provide guidance that is not only accurate, but also actionable, context-aware, and civically responsible.

In this work, we introduce Sage, a civic helpdesk designed to support both procedural understanding and strategic civic action, along with CivicBench, a benchmark for evaluating how well civic-minded AI systems provide this type of guidance. Together, they allow us to both build and systematically evaluate what effective, grounded, and responsible civic AI support should look like in real-world settings. Beyond this project, the benchmark plays an important role in moving the field forward by providing a more structured way to assess the quality of civic AI systems. In particular, it helps surface issues around accuracy, safety, and the real-world usefulness of model outputs in high-stakes situations. By making these evaluations more consistent and transparent, CivicBench contributes to the broader effort of ensuring that AI systems used in public-facing roles are reliable, accountable, and aligned with responsible civic engagement.

2. Related Work

Peter Levine's Civic Chatbot provides a strong foundation for AI-assisted civic engagement. It demonstrates how conversational systems can help users think through problems, ask better questions, and explore possible paths for civic action. By focusing on strategic guidance and supporting users in understanding their role within broader systems, it highlights the potential for AI to make civic knowledge more accessible and actionable.

Building on this foundation, there are opportunities to expand the system's capabilities to better support real-world use. In particular, incorporating local and real-time context, such as news or community-specific developments, can help connect guidance more directly to users' situations. Increasing transparency through citations and grounding responses in external sources can also strengthen trust, while developing more structured evaluation methods makes it possible to assess the quality of civic guidance in a consistent way. This project reflects those next steps, extending the original Civic Chatbot with a more grounded system design and a formal evaluation framework through CivicBench.

At the same time, existing benchmarks such as MMLU, HumanEval, and MedHELM have advanced the evaluation of large language models in areas like knowledge recall, code generation, and safety. While these benchmarks are valuable, they do not capture the type of reasoning required in civic contexts. Civic tasks involve understanding institutions, evaluating tradeoffs, and supporting real-world action, which are not well represented in traditional benchmark settings.

3. Sage's System Design

3.1 Overview

Sage is a civic helpdesk designed to support how people think through and act on complex civic problems, particularly in the context of wildfire recovery. Rather than simply providing answers, the system is built around the idea that civic engagement involves navigating uncertainty, understanding systems, and making decisions that often affect both individuals and communities.

At a high level, Sage focuses on helping users move from asking a question to figuring out what to do next. This includes not just providing information, but supporting reasoning, highlighting tradeoffs, and encouraging action in a way that is structured, grounded, and practical. The design of Sage reflects a broader goal of making civic guidance more accessible, usable, and responsive to real-world contexts, especially in situations where impact extends beyond a single individual.

3.2 Architecture

Sage is built on a modular architecture centered around a shared chatbot interface called CivicChatbot. This abstract class defines the basic structure that each chatbot-style component must follow. In the codebase, CivicChatbot requires each implementation to define two main behaviors: a response method, chat_with_bot(), and a logging method, log_bot(). This creates a consistent interface for components that may have different internal purposes but still need to receive input, generate a response, and record their output in a standardized way.

Sage, Ivy, and Orchid each inherit from this shared class, but they serve different roles in the overall system. Sage is the main user-facing civic helpdesk assistant. It receives user questions, retrieves relevant context when needed, generates responses through the LLM, and formats those responses for the web interface. Ivy acts as a local news retrieval component, using web scraping and summarization to identify recent, location-specific information that may be relevant to the user's situation. Orchid supports evaluation by acting as a simulated user or persona, generating prompts and participating in multi-turn conversations with Sage during benchmark testing. Because all three inherit from the same interface, they can be developed separately while still following a common interaction pattern.

This class structure is especially important for CivicBench because it helps standardize how different civic bots are evaluated. Any bot that implements the CivicChatbot interface can be passed into the benchmark pipeline using the same expected methods. In practice, this means

CivicBench does not need to be rewritten for every new model, agent, or system being tested. As long as a bot can respond through chat_with_bot() and record outputs through log_bot(), it can be evaluated in a consistent way. This makes the benchmark more flexible and reusable, allowing future systems to be compared against Sage using the same conversation and scoring process.

Beyond the shared class interface, Sage integrates several loosely connected components into one response pipeline. The Flask backend manages communication between the user interface and the chatbot. The LLM proxy handles response generation. The retrieval system provides grounding from curated civic and wildfire resources. Ivy optionally adds local news context when users enable that feature. Additional formatting and citation logic shapes the final response before it is returned to the frontend. This modular design allows each part of the system to be updated or improved independently while still working together as a unified civic helpdesk.

3.3 Sage System Design

RAG and Expert-Informed Mode

Our expert mentor Peter Levine of the Tufts Civic Studies Department was able to provide integral resources for this project. Our RAG setup starts with uploading civic studies and group organizing materials that Peter provided, including resources on civic participation, community organizing, coalition-building, and frameworks for thinking about collective action. These materials help Sage provide more thoughtful, strategic guidance, rather than just answering surface-level questions.

On top of this foundation, we added more specific wildfire-related resources, such as FEMA disaster assistance guides, SBA loan information, state and local government pages, and rebuilding and permitting resources. Together, this mix allows Sage to support both broader civic decision-making and more concrete, step-by-step recovery actions.

Prompting and Response Design

To make responses easier to follow and act on, Sage presents information in a clear and structured way. Instead of long, dense explanations, content is broken into bullet points, numbered steps, and short sections so users can quickly understand what to do, especially when dealing with more complex civic processes. This approach keeps the focus on clarity and usability, helping users scan responses and find what matters most.

At the same time, Sage emphasizes action. When appropriate, it suggests concrete next steps or offers to help draft materials such as emails, plans, meeting agendas, or checklists, making it easier for users to move from understanding a situation to taking action.

Interaction Style

When important context is missing, Sage does not immediately provide a full answer. Instead, it begins by asking a focused clarifying question to better understand the user's situation, such as their goals, constraints, or who is involved. This back and forth is intentionally limited, typically to one or two rounds of preliminary questions, after which Sage moves forward with the best possible response using the information available.

To make this process easier for users, Sage often presents these preliminary questions in a simple, structured format, such as:

- Immediate goal:
- Current situation:
- Key issues to address:
- Who's involved:

This format allows users to quickly copy and fill in the relevant details directly, making it clearer how they can move the conversation forward and provide the information Sage needs to give a more complete and useful response.

This interaction style was informed by findings from our user studies. While users valued the increased relevance and personalization that came from providing additional context, they also became frustrated when asked too many preliminary questions. As a result, Sage's design aims to balance these competing needs by keeping clarification brief while still gathering enough information to produce a more tailored and effective response.

Guardrails

Because wildfire recovery often involves high stakes and sensitive situations, Sage is designed to provide guidance in a careful and responsible way. It maintains a supportive and respectful tone without becoming overly emotional or making assumptions about the user's circumstances. Rather than presenting information as definitive or official advice, responses are framed as guidance and possible options. When details may vary by location or situation, Sage acknowledges this and encourages users to verify information with local sources.

At the same time, Sage stays within the scope of civic action, avoids suggesting harmful or unrealistic steps, and focuses on constructive ways users can move forward. This balance allows

it to remain both cautious and practical, offering meaningful support without overstating certainty or risk.

3.4 User Interface Features

Onboarding Flow

Sage includes a multi-step onboarding flow designed to introduce users to the system and set clear expectations before they begin interacting with it. This was especially important given the range of users we anticipate, including people with varying levels of technical experience. The goal was to make Sage approachable and easy to use for users of all ages and backgrounds.

The first screen, “Welcome to Sage,” introduces the tool and explains its purpose. It highlights that Sage can help with wildfire recovery and community action by guiding next steps, comparing approaches, and drafting practical materials. It also provides a simple overview of how to get started. The second screen introduces the response mode feature, which allows users to choose how Sage generates responses. The third screen introduces local news context, which allows Sage to incorporate more location-specific information into its answers.

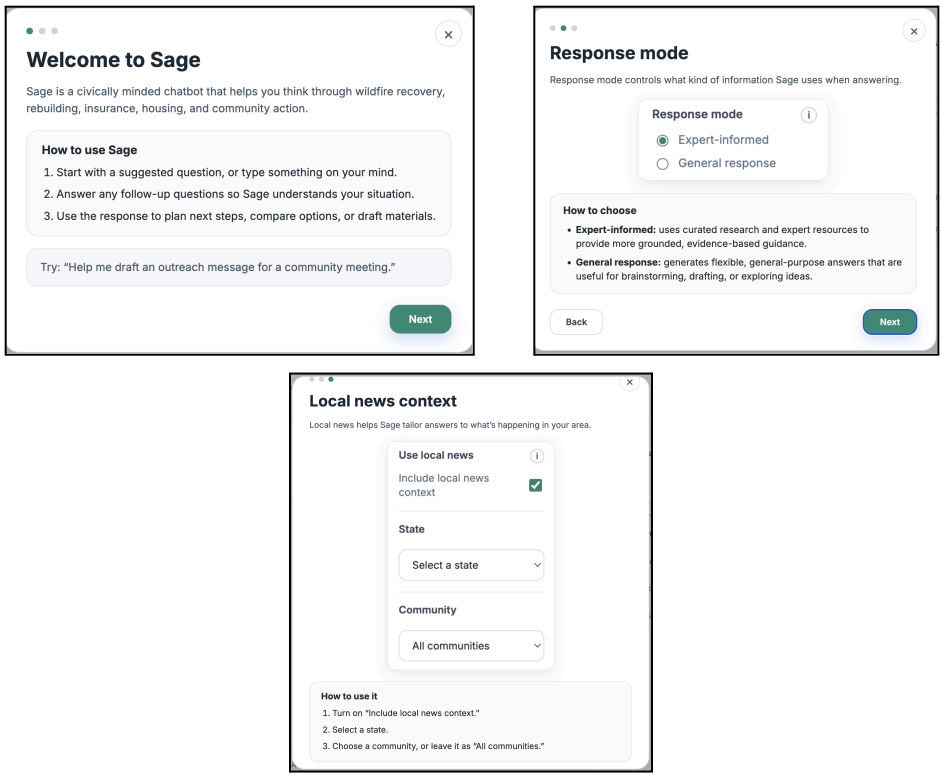


Figure 1: Sage onboarding flow introducing the tool, response mode options, and optional local news context.

Overall, the onboarding flow helps reduce confusion and builds user confidence by clearly explaining what Sage does and how to get started before the user begins interacting with the system.

Suggested Questions

Sage begins each interaction by introducing herself and providing a set of suggested questions. These are designed to help users quickly understand the types of support available, including both individual actions and group-based civic efforts. The initial suggestions are preselected to give users a starting point, especially if they are unsure how to phrase their question.

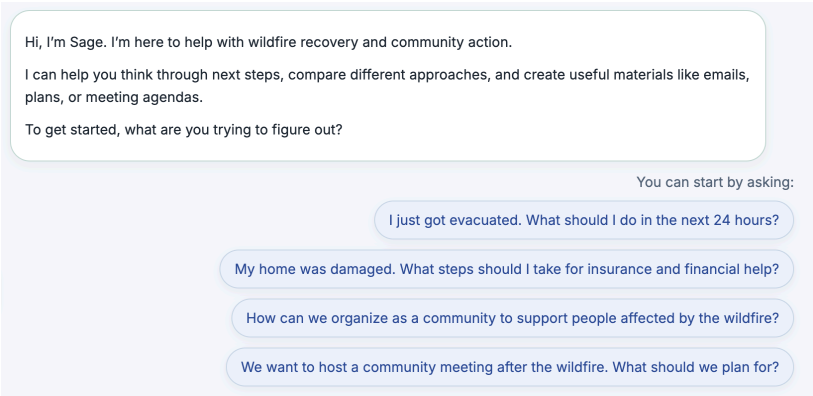


Figure 2: Sage's introductory message and suggested starting questions, which help users understand the types of civic support the system can provide.

As the conversation progresses, the suggested questions become more dynamic. Based on the topic and direction of the conversation, Sage anticipates what the user may want to explore next and offers relevant follow-up questions that can be selected with a click. This helps guide the interaction, encourages continued engagement, and makes it easier for users to move through more complex topics without needing to come up with each next step on their own.

Expert-Informed and General Response Toggle

The interface allows users to choose between expert-informed responses and general responses. Expert-informed mode uses the curated RAG materials to ground Sage's answers in civic studies and wildfire recovery resources. General mode allows users to receive a broader response without relying on those expert materials. This gives users more control over the type of guidance they want.

Local News Integration Through Ivy

Ivy adds a critical feature to Sage by bringing in local news context, allowing responses to reflect what is actually happening in a user's community in real time. In the civic studies space, this is especially important because effective civic action depends not only on understanding general processes, but also on being aware of current local conditions, decisions, and events. Access to timely, location-specific information helps users better understand what actions are possible, who is involved, and how situations are evolving on the ground.

This becomes even more valuable during active wildfire events, when information is constantly changing and people need quick access to updates about evacuations, relief efforts, local policies, and community responses. By integrating local news, Ivy helps bridge the gap between static guidance and real-world context, making Sage's responses more relevant, timely, and actionable when users need it most.

Technically, Ivy works by crawling local news sources from a database of outlets across the United States, organized by state and community using JSONL files. The original outlet directory was adapted from the Media and Democracy Project's journalism directory, which provided a broad starting point for identifying local news sources by region. When a user selects their location, Ivy retrieves the relevant outlets, crawls each homepage for articles it can parse, and uses an LLM to identify which articles are most relevant to the user's query. It then summarizes and returns key information, which Sage incorporates into its responses. To improve efficiency, Ivy caches results with timestamps, URLs, and summaries, allowing it to reuse recent data while still updating when information becomes stale.

Source Citations

Sage includes a source citation feature designed to make responses more transparent and trustworthy, while still remaining easy to use. Rather than presenting answers without context, Sage explains where its guidance comes from and how those sources informed the response. This is especially important in civic settings, where users are often making real-world decisions and need to understand the basis behind the advice they are given.

Instead of displaying a long, static list of sources, we designed the citation experience to be more intuitive and accessible. Each response includes a "Where did this information come from?" section that users can expand or minimize. When expanded, it first provides a short, plain-language explanation of how the sources shaped the guidance, followed by a list of the specific materials used. This approach helps users quickly understand the relevance of the sources before diving into the details.

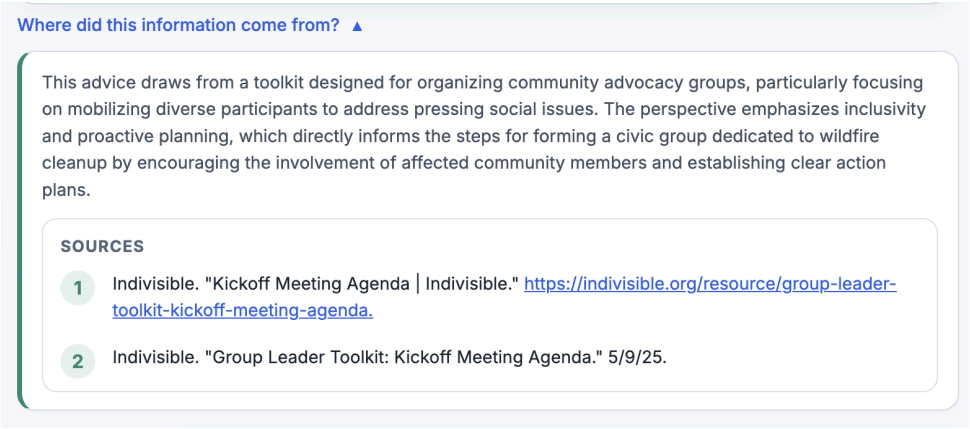


Figure 3: Example of Sage's expandable citation block, showing how the system explains where retrieved information came from and lists the sources used.

We intentionally avoided presenting only a traditional list of formal citations, as this can feel overwhelming or disconnected from the actual response. By combining a brief explanation with a collapsible list of sources, the design keeps the interface clean while still allowing users to explore the underlying materials when they want to. If none of the expert provided resources was used in generating a response a disclaimer is added to the end of the response that it is based on model trained information and not curated sources. This balance supports both usability and transparency, making it easier for users to trust and engage with Sage's guidance.

Drafting and Document Tools

Sage can help users draft practical materials such as emails, complaint drafts, meeting agendas, checklists, follow-up messages, and planning documents. Each response includes tools that allow users to edit, copy, or download the content as a Word document on the web application, making it easy to take what was discussed and use it directly outside the system.

This feature is especially important in a civic context because drafting documents is central to how individuals and groups organize and take action. Whether it is reaching out to neighbors, coordinating a meeting, contacting officials, or following up on next steps, much of civic engagement happens through written communication. By supporting this process directly, Sage moves beyond simply providing advice and helps users translate ideas into concrete action. At the same time, drafting supports accountability. Writing things down makes commitments more concrete and easier to follow through on, creating a shared reference point that helps ensure ideas turn into sustained action rather than stopping at discussion.

Including these tools also lowers the barrier to participation. Many users may know what they want to do but feel unsure about how to phrase it or where to start. By generating structured,

editable drafts, Sage helps users take that first step more confidently and supports the ongoing coordination that is necessary for effective civic engagement.

4. Benchmark

To evaluate Sage, we developed CivicBench, a benchmark designed specifically for civic AI systems. Unlike traditional benchmarks that focus mainly on knowledge recall, general reasoning, or code generation, CivicBench aims to measure how well a system can support real-world civic action, decision-making, and engagement. This is especially important because civic guidance often requires more than a correct answer. A strong response must identify relevant institutions, explain processes clearly, suggest realistic next steps, and encourage constructive participation.

4.1 Purpose and Scope

The purpose of CivicBench is to provide a structured way to evaluate whether an AI system can give useful and responsible civic guidance. Civic tasks are difficult to evaluate because they often combine factual, procedural, and strategic reasoning. Some questions have clear answers, such as which government body handles a specific decision. Other questions require judgment, such as how a neighborhood group should organize around a local problem or how a resident should respond when an official process is not working as expected.

CivicBench was designed to capture both types of tasks. It evaluates whether a system can answer procedural questions accurately, but also whether it can support broader civic reasoning. This includes understanding who has authority, what actions are available, what tradeoffs exist, and how users can move from confusion to practical action. In the context of Sage, this benchmark helps us evaluate wildfire recovery guidance not only as information support, but as civic support.

A broader goal of CivicBench is to make this evaluation reusable for other civic AI systems. Civic AI is a growing area, as more tools are being designed to help people understand institutions, access public information, draft civic materials, organize groups, and make decisions in public-facing contexts. Without a shared benchmark, it is difficult to compare systems or determine whether one civic bot gives better, safer, or more actionable guidance than another. CivicBench is intended to give future developers a standardized way to test their own civic AI bots against the same categories, questions, and rubrics.

At a high level, CivicBench takes a civic task and a bot-generated response or conversation, pairs it with the relevant rubric, and produces structured scores. These scores can be grouped by civic category and difficulty level, making the benchmark both evaluative and diagnostic. Rather than only producing an overall score, CivicBench helps identify where a civic AI system is strong or weak.

4.2 Civic Engagement Taxonomy

The foundation of CivicBench is a structured civic engagement taxonomy that defines the types of civic problems and interactions the system is designed to support. The taxonomy includes seven high-level categories: Civic and Political Knowledge, Civic Information Access and Media Literacy, Formal Participation in Government Processes, Civic Communication and Deliberation, Organizing and Associational Engagement, Community Problem-Solving and Public Work, and Relationships and Trust.

Each category is further broken down into five subcategories. These subcategories represent more specific civic skills or situations, such as identifying government authority, finding official documents, speaking at public meetings, writing to officials, starting a group, implementing a civic project, or building trust across networks. This structure allowed us to evaluate civic guidance at both a broad category level and a more specific task level.

To develop the taxonomy, we began with a research and synthesis process using ChatGPT to organize key themes across civic studies materials and examples of civic tasks. This allowed us to move efficiently while still grounding the structure in the expert-curated resources provided by Peter Levine. From there, Peter played a direct role in refining the taxonomy. He reviewed the categories, identified gaps, and contributed an additional category, helping ensure that the final structure better reflected how civic engagement happens in practice.

Rather than focusing on abstract or isolated questions, the taxonomy is designed to capture a broad range of civic life. It covers everything from understanding institutions, to finding trustworthy information, to participating in public processes, to organizing groups and sustaining relationships.

4.3 Task Design and Difficulty Levels

Using the taxonomy, we developed a standardized set of scenario-based questions for benchmark evaluation. The final benchmark set included 140 questions: four questions for each subcategory, five subcategories per category, across seven high-level civic categories. This gave

us broad coverage across different civic skills while keeping the evaluation set organized and interpretable.

The questions were designed to reflect realistic situations that users might bring to a civic helpdesk. Instead of asking only general questions like “What is local government?”, the benchmark includes applied prompts such as how to speak at a public meeting, how to verify a claim about a city decision, how to start a neighborhood group, or how to plan a community response after a wildfire.

Questions were initially generated with the help of ChatGPT, then reviewed by our team for quality, clarity, and fit with the intended subcategory. During review, we labeled weak generated questions when they were confusing, too vague, unrealistic, or did not match the intended civic skill. We then manually replaced those questions with clearer questions that preserved the original intent while better matching the benchmark’s goals. This review process helped ensure that the final question set was standardized enough for evaluation while still feeling realistic. Additionally, Peter Levine was able to validate these final sample questions.

Each question was also labeled by difficulty level: Easy, Lookup, or Hard. Easy questions test general civic knowledge that a model may already know from training data. Lookup questions require more specific or local information, making them useful for testing retrieval and grounding. Hard questions require multi-step reasoning, sequencing, or strategic judgment. This difficulty structure allows CivicBench to show not only whether a model performs well overall, but what kinds of civic tasks it handles reliably.

4.4 Rubric and Scoring Framework

To evaluate responses, we defined a set of core dimensions that reflect what strong civic guidance should include. These dimensions are Factual and Procedural Accuracy, Actionability, Contextual Relevance, Completeness, Clarity and Usability, and Civic Responsibility. Together, these dimensions allow CivicBench to measure both correctness and practical usefulness.

Factual and Procedural Accuracy evaluates whether the response correctly explains institutions, roles, rules, and processes. Actionability evaluates whether the response gives concrete steps the user can take. Contextual Relevance evaluates whether the answer is tailored to the user’s situation, location, and goals. Completeness evaluates whether the response covers the important parts of the task without leaving out major steps. Clarity and Usability evaluates whether the response is easy for a non-expert to understand and use. Civic Responsibility

evaluates whether the response encourages constructive, ethical, inclusive, and non-harmful civic engagement.

We then developed category-specific rubrics based on these dimensions. Each rubric contains multiple elements describing what a strong answer should include for that category. For example, a Civic and Political Knowledge response may be evaluated on whether it identifies the correct institution and explains the decision-making process. A Civic Communication response may be evaluated on whether it states a clear ask, uses respectful language, and addresses counterarguments. An Organizing and Associational Engagement response may be evaluated on whether it identifies stakeholders, proposes a strategy, and supports coalition-building.

Each rubric is scored out of 20 points. Elements can receive full credit, partial credit, no credit, or be marked not applicable when an element does not fit a specific prompt. This scoring structure keeps the benchmark interpretable while still allowing flexibility across different civic tasks. It also helps compare responses across categories because every category produces a normalized score, even though the specific rubric elements differ.

4.5 LLM-as-a-Jury and Orchid

CivicBench is designed to support LLM-as-a-jury evaluation at scale. In this setup, a civic bot's response is paired with the original benchmark question, its category, difficulty label, and relevant rubric. Multiple LLM judges can then evaluate the same response according to the same rubric. This reduces reliance on a single judge model and makes it possible to compare how different models assess the same civic guidance.

The benchmark was also designed to support multi-turn evaluation through Orchid, a persona agent that can simulate a user interacting with the civic bot. Orchid was included because different civic AI systems may handle user input differently. Some bots may answer directly, while others may require a back-and-forth conversation before producing a complete response. To make this fair and standardized, Orchid is prompted with a specific scenario and uses that same scenario when interacting with each bot being evaluated.

Orchid begins with the original benchmark question and can ask follow-up questions until it receives a satisfactory answer or reaches the conversation limit. In our design, the conversation ends after five turns if the answer has not been reached. This creates a consistent stopping point across evaluations, since a civic bot that cannot provide a usable answer within five turns should be scored lower for efficiency, completeness, or actionability.

CivicBench is also designed to support comparison across systems. In addition to evaluating Sage, the codebase includes a baseline bot, which we call a "raw bot." This provides a simpler

comparison point for measuring whether Sage's added system components improve the quality of civic guidance.

5. Evaluation Methods

5.1 Benchmark Evaluation

We evaluated Sage using the CivicBench framework described in Section 4. The goal was to test how well Sage performed across the benchmark's civic categories, subcategories, and difficulty levels.

For the final evaluation runs, we used the standardized benchmark questions as direct inputs to Sage. These questions were intentionally scenario-based and included enough context for Sage to answer without requiring an extended clarification phase. Because of this, the Orchid persona pipeline described in Section 4.5 was not central to the final Sage evaluation. Instead, each benchmark question was passed directly to Sage, and Sage's generated response became the main object of evaluation.

The final benchmark input was stored in eval/data/civic_judge.json. Each item included a question ID, a conversation field containing the prompt and Sage's response, a difficulty label, a high-level civic category, and a subcategory. The evaluation script then paired each response with the category-specific rubric from the rubric file.

Automated scoring was implemented through the FireBench judging script. For each item, the script built a judge prompt containing the conversation transcript and the relevant rubric rows. The judge was asked to assign an integer score for each rubric element, from zero up to that element's maximum point value, and return the result as a structured JSON object. The code also handled repeated rubric labels by adding numbered suffixes, allowing multiple rubric elements to belong to the same broader dimension while still being scored separately.

As described in Section 4.5, CivicBench supports an LLM-as-a-jury setup. In implementation, this means the same scoring process can be run with different judge models. The judge model is selected through a model argument, with GPT-4o-mini used as one judge option. The raw bot wrapper also supports other model options, including Claude Haiku, Gemini, Gemma, and Llama variants. Running the same benchmark responses through different judge models allows us to compare how different models evaluate the same civic guidance and reduces dependence on a single evaluator.

After scoring, the script extracted the structured score output and checked that each score fell within the allowed range for that rubric element. It then aggregated scores into normalized ratios, calculated as earned points divided by maximum possible points. This made results comparable across categories, even though each category used a different rubric. The evaluation also reported performance by difficulty level and high-level civic category, helping identify where Sage performed strongest and where it struggled.

This benchmark setup also allowed Sage to be compared with the raw bot baseline introduced in Section 4.5. In this baseline, prompts are sent directly to an underlying LLM without Sage's retrieval system, expert-informed prompting, guardrails, or response formatting. Comparing Sage to this baseline helps evaluate whether Sage's added system components improve the quality of civic guidance.

5.2 System-Level Evaluation

In addition to the automated benchmark evaluation, we conducted system-level evaluation to understand whether Sage worked reliably as a full civic helpdesk system. This included expert review, targeted retrieval testing, and informal testing of the hosted prototype.

First, we collected expert scores from Professor Peter Levine on a subset of Sage's benchmark responses. These responses were presented with their corresponding rubrics, and Peter assigned scores while also providing comments explaining why he scored them that way. This helped us understand not only whether Sage received high or low scores, but what specific civic reasoning, accuracy, or actionability issues influenced those scores. His feedback also gave us a way to compare automated LLM-judge scoring against expert civic judgment.

We also conducted targeted "needle in the haystack" evaluations to test whether Sage could retrieve specific information from the uploaded resource base. These tests asked Sage about details that should only be answerable if the retrieval system successfully found the relevant document or passage. This helped us evaluate whether the RAG pipeline was actually grounding answers in the curated materials rather than relying only on general model knowledge.

Finally, we tested Sage as a deployed prototype. The application was hosted on Render during the project so that team members, users, and Peter could interact with the live system rather than only testing it locally. Peter also explored the hosted site and provided feedback on the system's behavior, usability, and response quality. This helped us evaluate Sage as an end-to-end product, including the interface, response flow, citations, and drafting tools, rather than only evaluating isolated model outputs.

5.3 User-Level Evaluation

To evaluate how Sage performs in practice, we conducted a series of user studies focused on real interaction and decision-making. Participants were asked to behave as if they had experienced or were preparing for a wildfire situation and asked to use Sage as they naturally would while thinking out loud. This approach allowed us to observe not just what answers Sage produced, but how users interpreted those answers, what they found helpful or confusing, and whether they felt able to take action based on the guidance.

The scenarios covered a range of topics that combined both procedural and strategic elements. These included navigating FEMA assistance, dealing with delayed insurance claims, understanding housing and rebuilding options, organizing neighbors after a wildfire, and planning community-level recovery efforts. While participants did not necessarily have direct experience with these situations, the scenarios were designed to reflect realistic civic challenges that require both step-by-step guidance and higher-level reasoning about how to act.

We used a combination of qualitative and quantitative methods. During each session, participants were asked to think aloud as they interacted with Sage, which helped surface their reasoning process, moments of hesitation, and overall impressions in real time. After completing tasks, participants responded to Likert-scale questions evaluating clarity, usefulness, confidence in next steps, trust in the information, and overall effectiveness. We also asked open-ended follow-up questions to better understand how they would apply Sage's responses in a real-world setting.

Across studies, users consistently responded positively to structured outputs such as bullet points, step-by-step instructions, and clearly labeled sections. The ability to draft practical materials was also seen as a strong feature, reinforcing the idea that Sage functions as a tool for action rather than just a source of information. At the same time, feedback highlighted areas for improvement, particularly around keeping responses concise, incorporating relevant context, and balancing follow-up questions with responsiveness. Overall, these studies helped confirm that Sage is most effective when it supports users in moving from understanding a situation to taking concrete civic action.

5.4 Controlled Experiment: Interaction Style

We conducted a controlled experiment with eight participants from the Tufts MSIM 2026 cohort, representing a range of secondary disciplines including computer science, human factors, biomedical engineering, and product-focused fields. While participants did not have direct experience with wildfire recovery, this did not limit the value of the study. The goal of the

experiment was to evaluate Sage's interaction style and how effectively it supports general civic reasoning, rather than to test domain-specific wildfire expertise.

Participants spent 30 minutes and interacted with two versions of Sage: one that provided direct answers immediately, and another that began by asking a small number of clarifying questions before responding. The order of these conditions was randomized, and participants were not told which version they were interacting with at any given time. Each session included structured prompts based on civic scenarios, followed by more open-ended exploration, allowing participants to engage with the system more naturally.

Throughout the sessions, participants were encouraged to think aloud, which provided insight into how they processed the interaction and where friction or confusion occurred. We measured both behavioral and subjective outcomes. Behaviorally, we observed factors such as interaction length, number of turns, and whether users engaged with follow-up suggestions or drafting tools. Subjectively, participants rated each experience using Likert-scale questions on clarity, usefulness, trust, efficiency, and overall satisfaction, and provided qualitative feedback on their preferences.

The results showed that the question-first interaction style generally led to more contextually relevant and tailored responses, particularly for more complex or strategic prompts. However, it also introduced friction when the clarification phase was too long or felt unnecessary. Participants appreciated when Sage asked focused questions that clearly improved the response, but became frustrated when the interaction required too much back-and-forth before receiving guidance.

These findings directly informed Sage's final interaction design. Rather than choosing one approach over the other, we adopted a hybrid model that limits the clarification phase to one or two rounds of targeted questions. This allows Sage to gather enough context to improve response quality while maintaining a smooth and efficient user experience.

6. Results Analysis

The results from our evaluation show both the promise and the difficulty of building civic AI systems. Across the benchmark, Sage produced responses that were generally strong, especially for tasks involving clear civic procedures, formal participation, and structured next steps. However, the results also show that performance varied depending on the type of civic task, the judge model used, and whether the evaluation prioritized completeness, concision, or civic judgment.

Three main findings emerged from the results. First, LLM judges did not score Sage responses identically, even when using the same CivicBench rubrics. This supports the need for a jury-style evaluation process rather than relying on a single automated judge. Second, expert review showed that Sage was strongest on procedural civic tasks but weaker on organizing, local variation, and informal civic relationships. Third, the comparison between Sage and a raw baseline bot revealed that longer responses may receive higher automated scores, even when shorter and more focused responses may be more usable for real users.

These findings are important because civic guidance is not only about whether a response is factually correct. A strong civic response also needs to help users understand where authority sits, what actions are available, which relationships matter, and how to act responsibly in a specific context. The results therefore help evaluate Sage not only as a chatbot, but as a civic support tool.

Together, these findings suggest that Sage has a strong foundation as a civic helpdesk, but also that evaluating civic AI requires more than measuring whether an answer sounds complete. Strong civic guidance must be accurate, actionable, locally aware, responsible, and usable.

6.1 Benchmark Results Across Judge Models

The first part of our evaluation compared how different LLM judge models scored the same set of Sage-generated benchmark responses. Each response was scored using the category-specific CivicBench rubric, and the final score was normalized as the ratio of earned points to maximum possible points. This normalization made the results comparable across civic categories, even though each category had its own rubric criteria.

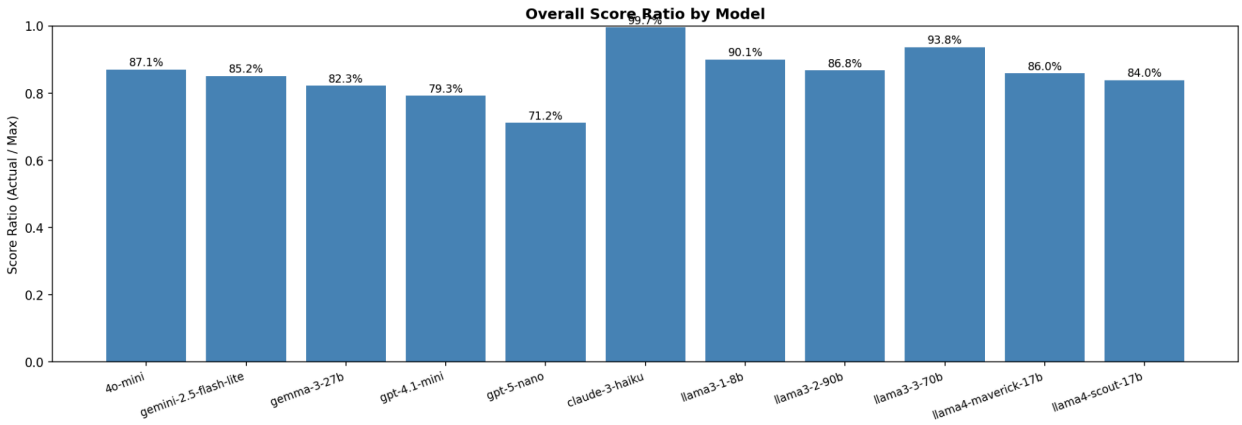


Figure 4: Overall normalized CivicBench score ratios assigned by each LLM judge model, showing variation in how different models applied the same rubrics to the same Sage responses.

Across all tasks, judge models assigned noticeably different overall scores. Claude Haiku gave the highest average score, with an overall normalized score of 99.7%, while GPT-5-nano gave the lowest average score, with an overall normalized score of 71.2%. Other judge models fell between these two extremes. This variation is important because all models were judging the same Sage-generated responses using the same rubrics. The difference therefore reflects variation in how judge models applied the rubric, not variation in the underlying chatbot responses.

One notable finding is that the judge model did not necessarily favor responses generated by a model from the same provider or model family. The benchmark transcripts were generated by Sage using GPT-4o-mini, but GPT-based judges did not consistently assign the highest scores. In fact, GPT-5-nano gave the lowest overall scores, and GPT-4o-mini rated the responses lower than several non-GPT models, including Claude Haiku, Llama-3.1, and Llama-3.3. This suggests that model-origin bias was not the main explanation for the scoring differences.

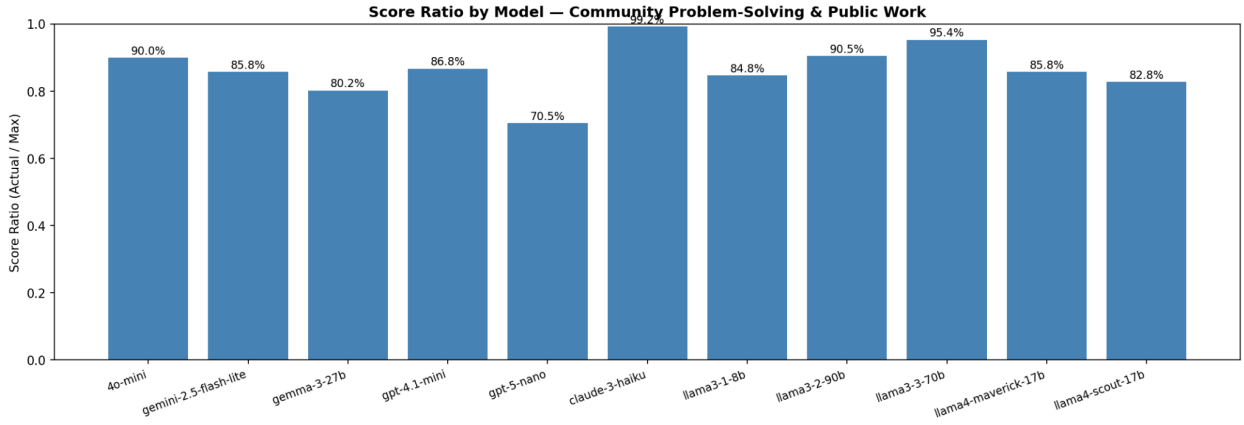


Figure 5: Example category-level score ratios by judge model for Community Problem-Solving and Public Work tasks.

The category-level benchmark charts also showed that judge behavior varied by civic task type. For example, Gemma gave relatively low scores on Organizing and Associational Engagement tasks, but gave higher scores on Civic and Political Knowledge tasks. This suggests that judge models may differ not only in overall strictness, but also in how they interpret different kinds of civic reasoning. A model that is generous on procedural knowledge may be stricter on organizing strategy, relationship-building, or civic responsibility.

This matters because CivicBench is designed to evaluate open-ended civic guidance, not multiple-choice answers. In open-ended evaluation, there is often more than one reasonable way to judge quality. Some judges may reward completeness, while others may prioritize

precision, caution, or practical actionability. The differences across judge models show that automated civic evaluation is itself a judgment task.

This finding supports CivicBench's LLM-as-a-jury design. Since different judge models apply the same rubric differently, relying on a single judge could produce a misleading picture of system performance. A multi-model judging setup provides a broader and more stable view of how a civic AI system performs across task types. It also makes the benchmark more transparent because it allows us to see where judge models agree and where evaluation is more uncertain.

6.2 Expert Evaluation Results

To evaluate Sage's civic reasoning more directly, we asked Professor Peter Levine, an expert in civic studies, to score a subset of Sage's benchmark responses using our civic engagement taxonomy rubrics. This expert review was important because civic guidance cannot be evaluated only by surface-level correctness. A response may be factually accurate but still miss important elements of civic action, such as local variation, power, accountability, trust, or strategic leverage.

Across the expert-evaluated responses, Sage earned 111 out of 140 points, for an overall expert score of 79.3%. This suggests that Sage performed well as a civic guidance tool, but its effectiveness depended on the type of civic task. Sage was strongest on structured civic processes, especially the task about speaking at a city council meeting, where it received a perfect score. In that case, Sage clearly explained the venue, rules, deadlines, action steps, evidence use, and follow-up. This shows that Sage is especially effective when the civic process has a clear sequence and the user needs help understanding what to do before, during, and after participation.

Sage also performed well on tasks involving planning, networks, and community problem-solving. These responses showed that Sage can help users organize their thinking, identify relevant actors, and consider concrete next steps. However, Peter's feedback also showed that some of these otherwise strong answers needed more attention to civic values such as safety, equity, accountability, honesty, boundaries, and trust repair. This suggests that Sage can generate useful civic plans, but sometimes needs stronger grounding in the ethical and relational responsibilities of civic work.

The expert review also showed that Sage's responses were sometimes too general. In tasks involving local government, civic information access, and organizing, Sage often provided useful broad guidance but did not always give enough local specificity or identify clear civic leverage

points. This matters because civic authority varies by jurisdiction, and users often need to know not only what institutions do, but how they can actually influence them.

Sage's weakest expert-evaluated area was Organizing and Associational Engagement. In this case, the response relied too much on online research and not enough on talking to people, understanding group structure, navigating power dynamics, and building leadership. This was an important finding because organizing is not just an information task. It depends on relationships, trust, shared leadership, and understanding how groups actually work.

Overall, the expert evaluation suggests that Sage has a strong foundation, especially for making formal civic processes easier to understand. However, it needs further work when civic action depends on local context, informal networks, organizing strategy, and relational judgment. We do not interpret this as a failure, but as a useful learning outcome: effective civic chatbots need to go beyond explaining procedures and help users understand where power sits, who to talk to, and how to act responsibly within a specific community.

6.3 Alignment Between LLM Judges and Expert Judgment

To better understand whether automated LLM judges could approximate expert civic judgment, we compared AI-generated rubric scores with Professor Levine's expert scores. This analysis used the seven expert-evaluated questions, each scored across six rubric criteria, for a total of 42 rubric-question data points.

We used two complementary similarity measures: normalized MAE similarity and Pearson correlation. These metrics were included because they capture different kinds of agreement between an AI judge and an expert human evaluator.

The first metric, normalized MAE similarity, measures how close each AI judge's scores were to the expert's scores. Since different rubric criteria had different maximum point values, we normalized each score by its criterion-specific maximum before comparing them. This made it possible to compare scoring differences across rubric elements worth different numbers of points. We then calculated mean absolute error and reported similarity as 1 - MAE. A value closer to 1 means that the AI judge's scores were closer to the expert's scores.

The second metric, Pearson correlation, measures whether the AI judge followed a similar scoring pattern to the expert across questions and rubric dimensions. This is different from exact score closeness. A model may be stricter or more generous than the expert overall, but still recognize the same responses as stronger or weaker. Pearson correlation captures this kind of relative agreement.

Together, these two metrics give a fuller picture of judge alignment. Normalized MAE similarity asks whether the AI judge gave scores close to the expert's scores. Pearson correlation asks whether the AI judge identified a similar pattern of strengths and weaknesses. Both are important for CivicBench because the benchmark is meant not only to produce scores, but also to diagnose where a civic AI system performs well or poorly.

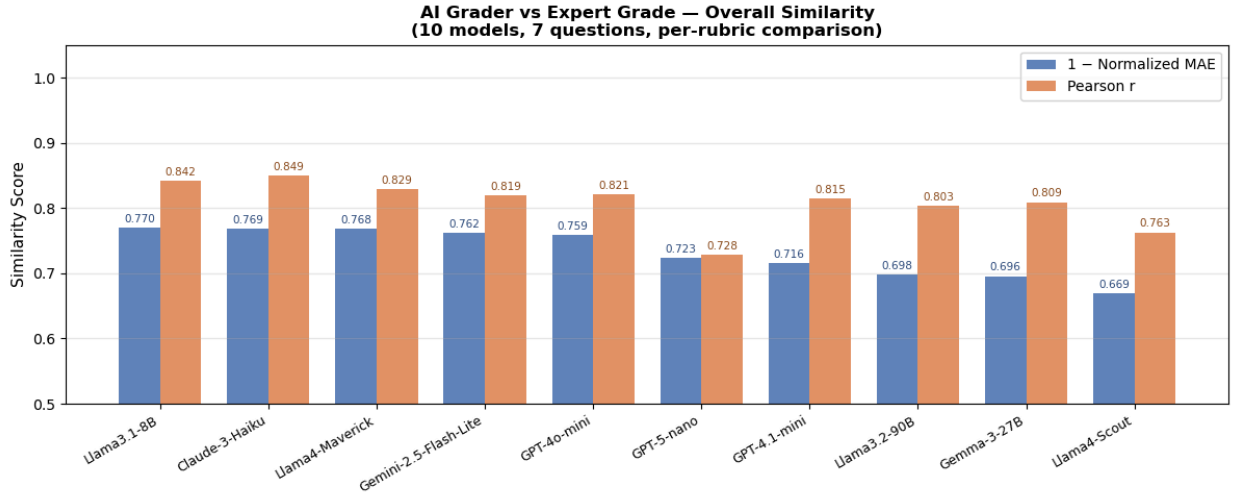


Figure 6: Similarity between LLM judge scores and expert scores across seven expert-evaluated questions, showing both exact score closeness through normalized MAE similarity and relative scoring-pattern agreement through Pearson correlation.

The strongest models by normalized MAE similarity were Llama3.1-8B at 0.770, Claude-3-Haiku at 0.769, and Llama4-Maverick at 0.768. These values were very close to one another, suggesting that several judge models reached a similar level of agreement with the expert under the current rubric and prompting setup. There was no single model that clearly dominated the others by exact score closeness.

Claude-3-Haiku had the highest Pearson correlation, with a score of 0.849. This means that Claude-3-Haiku most closely matched the expert's relative scoring pattern across the evaluated questions and rubric criteria. In other words, it was especially strong at identifying the same kinds of responses and rubric dimensions as stronger or weaker, even if its exact score values were not always identical to the expert's.

At the lower end, Llama4-Scout had the weakest normalized MAE similarity, with a score of 0.669, and a Pearson correlation of 0.763. GPT-5-nano also showed weaker alignment, with an MAE similarity of 0.723 and Pearson correlation of 0.728. This suggests that GPT-5-nano differed from the expert both in absolute score values and in how it ranked the relative strength of responses.

Across the ten models included in this comparison, normalized MAE similarity scores clustered between approximately 0.67 and 0.77, while Pearson correlations ranged from approximately 0.73 to 0.85. The fact that Pearson correlations were generally higher than MAE similarity scores suggests that AI judges were better at recognizing the overall pattern of expert judgment than matching exact expert scores. In practical terms, this means LLM judges may be useful for identifying relative strengths and weaknesses, but their numeric scores should still be interpreted cautiously.

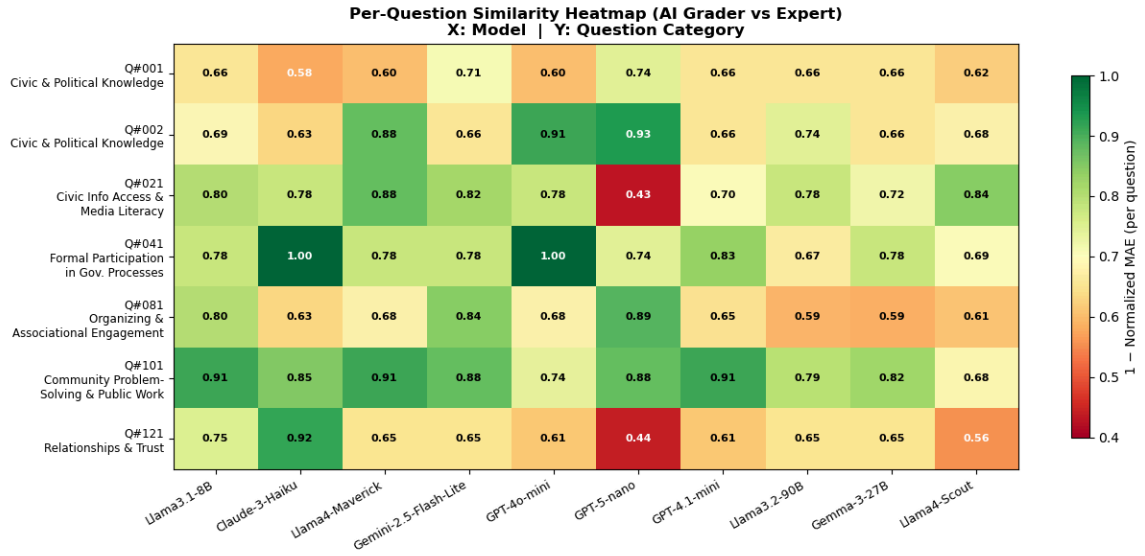


Figure 7: Per-question similarity heatmap showing that no single LLM judge aligned best with the expert across every civic task category.

The per-question similarity heatmap added another layer to this result. No single model aligned best with the expert across every question category. Some models aligned more closely on civic knowledge tasks, while others aligned better on relational or organizing-oriented tasks. This suggests that civic evaluation may not have one universally best automated judge. Different models may be better calibrated for different types of civic reasoning.

However, this category-level pattern should be interpreted carefully because the expert comparison was based on a limited subset of questions, with only one or two examples per category. The results are not enough to make statistically strong claims about which model is best for each civic category. Instead, they suggest a useful direction for future work: CivicBench could eventually explore category-specific judge selection or model routing, but this would require a larger expert-scored dataset.

The AI-generated grading explanations also provided useful evidence that the rubrics were working as intended. In lower-scoring categories, several Llama-family models independently penalized responses for similar missing civic competencies. For example, on the Organizing and

Associational Engagement task, they noted missing guidance on internal conflict, decision-making processes, and leadership development. On the Relationships and Trust task, they flagged missing attention to reciprocity, follow-through, marginalized groups, conflict resolution, and trust repair.

This consistency suggests that the rubric was not only producing numbers, but also guiding models toward meaningful civic evaluation criteria. The judge models were not simply giving vague impressions of answer quality. They were often identifying specific missing dimensions of civic action. At the same time, because their exact scores still differed from the expert and from each other, the results reinforce the need to combine automated judging with expert review.

6.4 Sage Compared to a Raw Baseline Bot

We also compared Sage against a raw baseline chatbot. The baseline sent the same prompts directly to an underlying model without Sage's RAG database, expert-informed prompting, response formatting, citations, or safety guardrails. This comparison was designed to test whether Sage's added system components improved the quality of civic guidance.

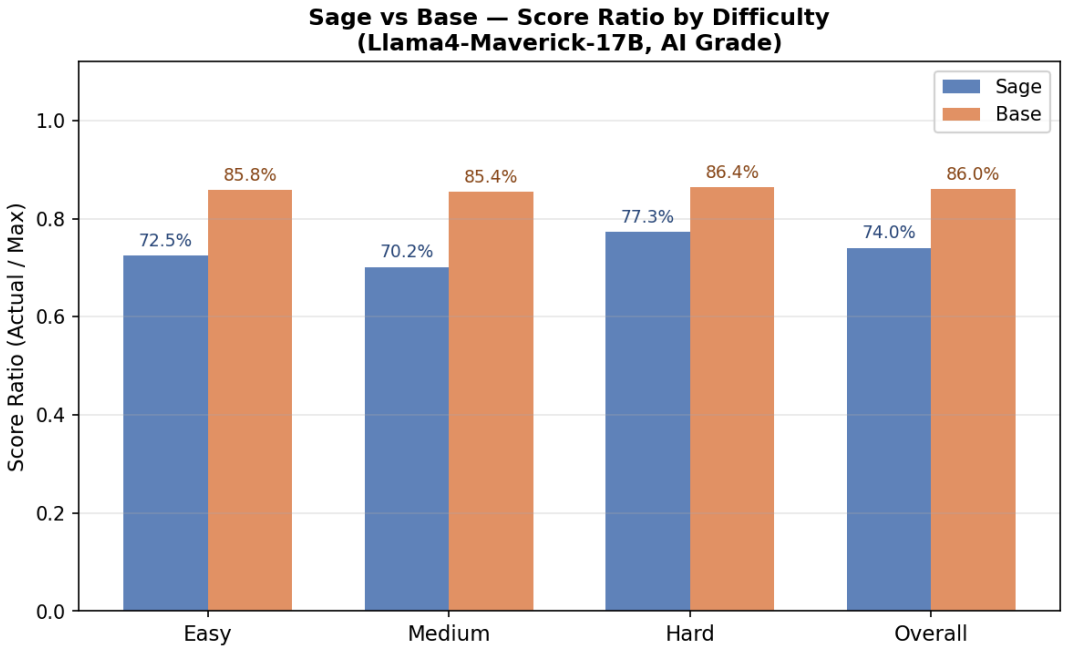


Figure 8: Automated score comparison between Sage and the raw baseline bot by difficulty level, showing that the baseline received higher LLM-judge scores across Easy, Medium, Hard, and overall categories.

In the automated grading results, the raw baseline scored higher than Sage across difficulty levels. Sage received 72.5% on Easy questions, 70.2% on Medium questions, 77.5% on Hard

questions, and 74.0% overall. The baseline received 85.8% on Easy questions, 85.4% on Medium questions, 86.4% on Hard questions, and 86.0% overall. At first, this appears to suggest that the baseline performed better than Sage.

However, a closer comparison of response length revealed an important explanation. Sage's responses were much shorter than the baseline responses across all difficulty levels. On average, Sage produced about 203 tokens per response, while the baseline produced about 477 tokens per response.

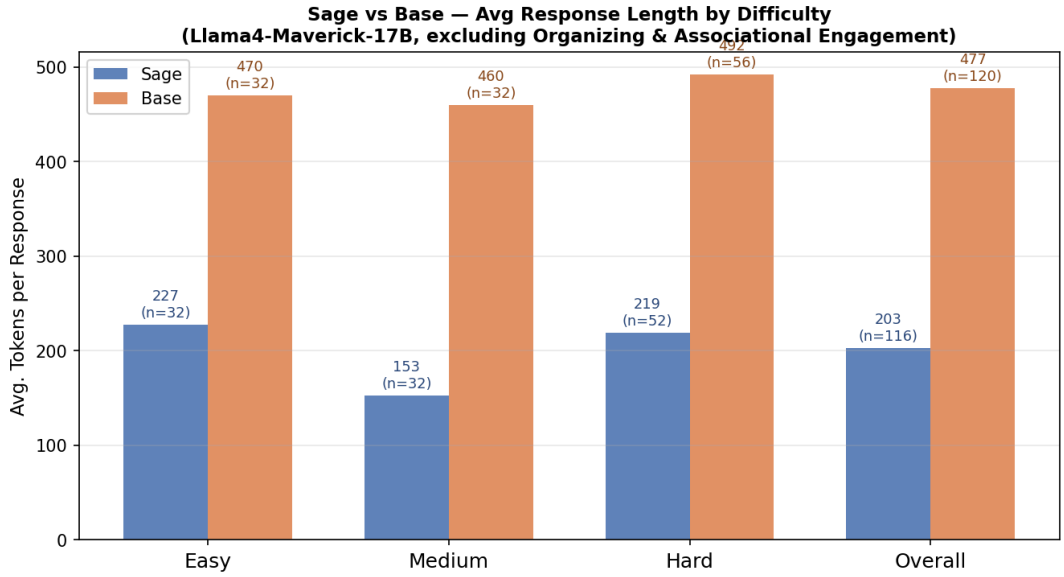


Figure 9: Average response length for Sage and the raw baseline bot by difficulty level, showing that the baseline produced much longer responses across all task types.

For Easy questions, Sage averaged 227 tokens compared to the baseline's 470 tokens. For Medium questions, Sage averaged 153 tokens compared to the baseline's 460 tokens. For Hard questions, Sage averaged 219 tokens compared to the baseline's 492 tokens.

This response-length difference likely affected the automated scores. Because CivicBench rubrics reward coverage across multiple dimensions, longer responses have more opportunities to mention relevant procedures, stakeholders, risks, follow-up steps, and civic values. A longer response may therefore satisfy more rubric elements, even if it is less concise or more overwhelming for a real user.

This result does not necessarily mean the raw baseline was a better civic assistant. Instead, it reveals an important evaluation challenge. Automated judges may reward comprehensiveness more than practical usability. Sage was intentionally designed to produce shorter, more structured responses that reduce user burden and make guidance easier to act on. This design

choice was also supported by user feedback and expert feedback, which suggested that concise but complete responses were often preferable.

The comparison is especially important because Sage's design goals were not identical to the behavior rewarded by the automated rubric. Sage was built to be grounded, careful, and usable. It uses retrieved materials and guardrails to narrow its response, avoid overstating certainty, and keep answers focused. The baseline, by contrast, was freer to generate broad and expansive answers. Those answers may cover more rubric items simply because they include more content, but that does not automatically make them better for a user trying to make a decision under stress.

The Sage/base comparison therefore highlights a tension between rubric coverage and user-centered design. A response that scores well on an automated rubric may not always be the most useful response in a real civic setting, especially if the user is stressed, overwhelmed, or looking for clear next steps. Future versions of CivicBench should account more directly for concision, prioritization, and cognitive load, rather than rewarding length alone.

One possible improvement would be to add a rubric element that evaluates whether the response is appropriately scoped. This would reward answers that are complete enough to be useful without overwhelming the user. Another improvement would be to distinguish between "mentions many relevant ideas" and "prioritizes the most important next steps." This distinction matters in civic contexts because users often need help deciding what to do first, not only a long list of everything they could possibly do.

6.5 Summary of Results

Taken together, the results show that Sage has a strong foundation as a civic helpdesk, but that its strengths and weaknesses depend on the type of civic task being evaluated. Sage performed best on structured procedural tasks, especially when the user needed help understanding a formal process or preparing for a specific civic action. Its perfect expert score on the city council meeting task shows that Sage can make formal participation more approachable when the process is clearly defined.

The results also show that Sage needs further improvement on tasks that require local context, organizing strategy, and relational judgment. Expert feedback showed that Sage sometimes gave generally helpful answers without fully identifying civic leverage points, local variation, power dynamics, accountability, or trust-building strategies. These are not minor details in civic work. They are often the difference between knowing about a process and knowing how to act effectively within it.

The evaluation also demonstrates the value of CivicBench as a diagnostic framework. LLM judges made scalable evaluation possible, but the variation across judge models showed why a jury-style approach is important. The expert comparison showed that automated judges can approximate some patterns of human civic judgment, but not perfectly. The Sage/base comparison revealed that automated rubrics may reward longer answers, even when users may prefer shorter and more focused guidance. Finally, the user study results helped connect these evaluation findings back to real interaction design, showing why Sage needs to balance personalization, completeness, and efficiency.

Overall, these findings suggest that building civic AI requires both strong system design and careful evaluation design. Sage shows that AI can help users move from confusion to concrete civic action, but the results also show where future work is needed: stronger local grounding, better support for organizing and relationships, clearer civic leverage points, and evaluation methods that balance completeness with usability.

More broadly, the results reinforce the central argument of the project. Civic AI systems should not only be judged by whether they provide polished or informative responses. They should be judged by whether they help people understand systems, identify realistic paths for action, work with others, and participate responsibly in civic life. Sage makes progress toward this goal, and CivicBench provides a framework for measuring that progress more systematically.

7. Limitations

Sage and CivicBench are still research prototypes, and several limitations shaped the scope of our results. These limitations do not undermine the evaluation, but they do affect how the findings should be interpreted. Overall, the results are best understood as diagnostic rather than definitive.

First, our user studies did not include participants who had personally experienced wildfire recovery or who had extensive professional expertise in wildfire response, insurance, emergency management, or rebuilding. This means our evaluation was useful for testing clarity, usability, interaction style, and general civic reasoning, but less conclusive for measuring how Sage would perform for users facing a real disaster. Since wildfire recovery involves stress, urgency, uncertainty, and material consequences, future testing with affected residents and domain professionals would be necessary before treating Sage as ready for real-world deployment.

Our participant pool was also limited in size and background. Most participants were students or university-affiliated users, which gave us helpful feedback from people familiar with digital tools, but may not fully represent the needs of older adults, displaced residents, low-connectivity users, non-native English speakers, or people under immediate stress after a wildfire. This limitation matters because a civic helpdesk should be accessible to people with different levels of technical comfort, time, trust, and emotional capacity.

There are also technical limitations. Sage depends on the quality and coverage of its RAG materials, so gaps in the uploaded resources can lead to incomplete or overly general answers. This appeared in the results when Sage sometimes gave broadly useful civic guidance but did not provide enough local specificity or civic leverage. Ivy helps add local news context, but we did not enable local news retrieval during benchmark evaluation because crawling and summarizing local articles would have required too much time and too many model calls across the full question set. As a result, the benchmark primarily evaluated Sage's curated resource grounding and response design, not the full local-news-enabled version of the system.

The benchmark itself also has limitations. CivicBench relies on LLM judges, which makes evaluation scalable but not perfectly objective. Our results showed meaningful differences across judge models, suggesting that automated evaluation is sensitive to which model is used as the judge. Expert review helped address this, but only on a subset of responses. This means the results should be interpreted as diagnostic rather than definitive. They are useful for identifying patterns, strengths, and weaknesses, but not for making final claims about Sage's performance in all civic contexts.

The Sage and baseline comparison also revealed an evaluation limitation: automated rubrics may reward longer, more comprehensive responses even when shorter responses are more usable. Future evaluation should therefore account for prioritization, cognitive load, and appropriate scope.

Finally, Sage provides guidance, not official legal, financial, insurance, or emergency advice. In high-stakes wildfire recovery contexts, users still need to verify deadlines, eligibility, permits, claims information, and emergency instructions with official agencies or qualified professionals. Sage can help users understand systems and prepare for action, but it should not replace authoritative sources or human expertise.

8. Future Work

The results of this project point to several directions for future work. Rather than treating Sage's weaknesses as a failure, we see them as evidence of what civic chatbots need in order to become more useful, responsible, and realistic. Sage showed promise in structured civic tasks, but future versions should better support local context, organizing, relationships, and user-centered evaluation.

For Sage, the most important next step is making the system more locally actionable. Many of Sage's lower-scoring responses were not incorrect, but they were too general. Future versions should ask more targeted location-based questions and connect users to specific institutions, agencies, public meetings, documents, deadlines, and decision points in their community. This would help Sage move from explaining how civic systems usually work to helping users understand how their own local system works.

Sage should also provide clearer civic leverage points. Expert feedback showed that Sage sometimes explained what government bodies do without fully explaining how residents can influence them. Future versions should better identify who has authority, when public input matters, what evidence is useful, and which actions are most likely to make a difference.

Another priority is stronger support for organizing and relationship-building. Sage's weakest expert-evaluated performance was in Organizing and Associational Engagement, where it relied too much on online materials and not enough on talking to people, understanding group structure, and navigating power dynamics. Future versions should better guide users through building trust, finding local leaders, handling disagreement, developing leadership, and sustaining participation over time.

For CivicBench, future work should expand the benchmark and continue validating it with experts. This includes testing more civic domains, improving judge prompts, comparing more base models, and increasing the number of expert-scored examples. A larger expert-scored dataset would help determine which LLM judges align most closely with human civic judgment and whether different judge models are better suited for different civic categories.

CivicBench should also improve how it evaluates usability. The Sage/base comparison showed that longer responses may receive higher automated scores because they cover more rubric elements. Future rubrics should better distinguish between responses that mention many relevant ideas and responses that prioritize the most useful next steps. Strong civic guidance should be complete, but also concise, clear, and manageable for users.

Finally, future evaluation should include larger and more representative user testing. Future studies should involve people with wildfire experience, emergency response professionals, housing advocates, insurance experts, local officials, community organizers, older adults,

non-native English speakers, and users with different levels of technical comfort. This would help determine whether Sage's guidance is not only clear and usable, but accurate, trustworthy, and actionable in real recovery situations.

9. Conclusion

This project explored how generative AI can support civic action in the context of wildfire recovery. Sage was designed as a grounded civic helpdesk that helps users understand systems, compare options, organize with others, and draft practical materials. Rather than treating wildfire recovery as only an information problem, Sage approaches it as a civic problem involving institutions, relationships, strategy, and collective action.

The results show that Sage has a strong foundation, especially for structured civic tasks such as explaining formal processes and helping users identify concrete next steps. At the same time, the evaluation revealed important areas for improvement. Sage was less effective when civic action required local specificity, informal networks, organizing strategy, power analysis, and relationship-building. We do not view these weaknesses as a loss, but as one of the most valuable outcomes of the project. They clarify what future civic chatbots need in order to become more useful, responsible, and realistic.

CivicBench also proved valuable as a diagnostic framework. By combining a civic taxonomy, scenario-based questions, category-specific rubrics, LLM-as-a-jury scoring, and expert feedback, it helped show not only how Sage performed, but where and why it struggled. The results also showed that civic AI evaluation must balance completeness with usability, since longer answers may score well while shorter, more focused answers may better serve real users.

Together, Sage and CivicBench show that building civic AI requires both careful system design and careful evaluation design. A useful civic assistant must be grounded, transparent, practical, locally aware, and tested against standards that reflect real civic life. While Sage remains a prototype, this project offers a foundation for future civic AI systems that help people move from uncertainty to informed, constructive action.

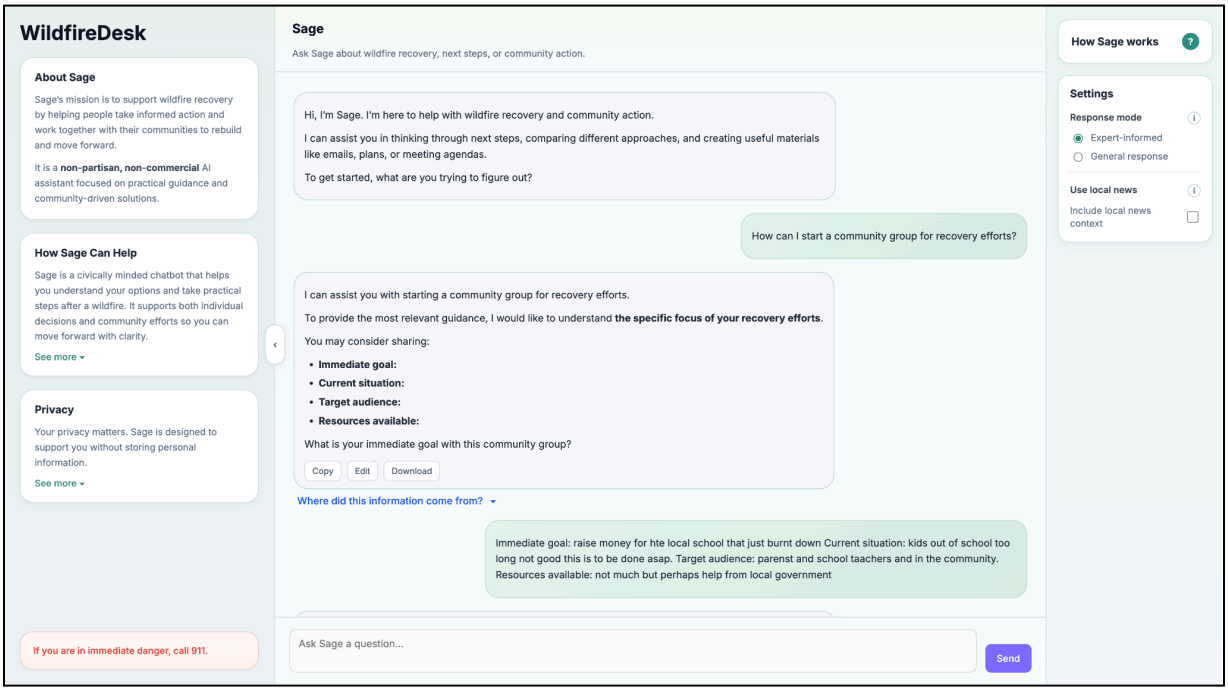
10. Appendix

Appendix A: Hosted Sage Prototype

We are temporarily hosting Sage on Render so that users, team members, and our expert advisor could interact with the live prototype.

Hosted prototype: https://wildfiredesk-firebench.onrender.com/

Appendix B: Sage Interface Screenshot



Appendix C: Civic Engagement Taxonomy Example

1. Civic & Political Knowledge		Understanding how the government works, who makes decisions, and what public issues mean. This enables informed participation.	
Subcategory		**Example Question (there will be 3-4 each)**	**Complexity**
a	Roles and authority in local government	In a typical U.S. city/town, what do the city council/select board, planning board, and zoning board of appeals do, and how do their roles differ?	Easy (already known)
b	Policy Terminology	What is the difference between a zoning variance, a special permit, and a rezoning? Give a plain-language explanation with examples.	Easy (already known)
c	Jurisdiction Specific authority	In [CITY], who has final authority to approve zoning changes, and where is that	Lookup (needs retrieval)

		authority described (charter/ordinance/bylaws)?	
d	Process mental model	A developer proposes housing on land currently zoned industrial. What steps are likely required (e.g., rezoning, special permit, site plan review), and which bodies are typically involved in what order?	Hard (multi step reasoning)
e	Budget & policy cycle literacy	I want to influence the annual budget in [CITY]. What is the usual timeline (proposal → hearings → amendments → adoption), and at which points is resident input most influential?	Hard (multi step reasoning)

Our full Civic Taxonomy can be found here.

Appendix D: Rubric Example

Rubric #1: Civic & Political Knowledge

Label	Element	Description	Why it matters	Points
Factual & Procedural Accuracy	The bot correctly identifies institutional roles and authority	The bot accurately names the relevant government body and explains what it can or cannot decide	This matters because effective civic engagement depends on correctly identifying where decision-making power lies.	5
Factual & Procedural Accuracy	The bot accurately explains the process sequence	The bot describes the correct step-by-step flow of how decisions are made	This matters because understanding how decisions move through a system is necessary for meaningful participation.	5
Clarity & Usability	The bot clearly defines key terminology	The bot explains civic terms in plain language and distinguishes similar concepts	This matters because users cannot engage effectively if they do not understand key civic concepts.	3

Contextual Relevance	The bot grounds the answer in the correct jurisdiction	The bot references relevant local laws, charters, or governing structures when needed	This matters because civic processes vary by location, and incorrect generalization leads to unusable advice.	3
Actionability	The bot identifies civic leverage points	The bot highlights where and how individuals can influence the process	This matters because knowledge should translate into actionable opportunities for participation.	3
Civic Responsibility	The bot appropriately expresses uncertainty or variation	The bot acknowledges when rules vary or information may depend on location	This matters because overconfident or incorrect certainty can mislead users in real civic contexts.	1

Our full Rubrics can be found here.

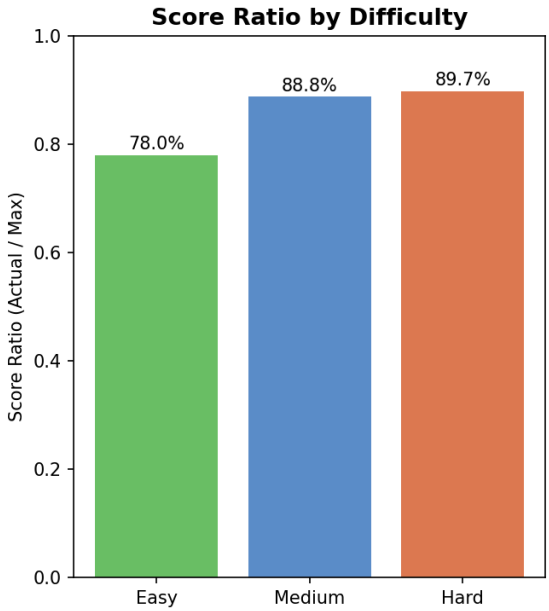
Appendix E: Evaluation Questions Examples

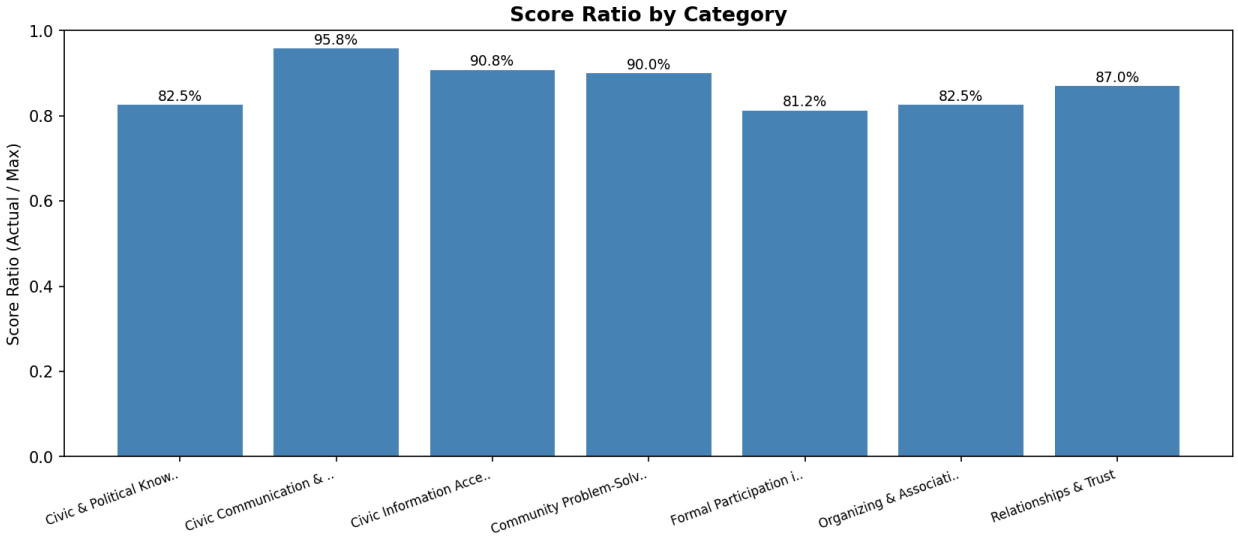
Category	Subcategory	Scenario	Evaluation Question
Civic & Political Knowledge	Roles and authority in local government	A group of residents is trying to stop a new housing development, but they are unsure which local government body actually has the authority to approve or deny the project.	Explain the roles of the city council/select board, planning board, and zoning board of appeals, and how their responsibilities differ in a typical U.S. city or town.
Civic & Political Knowledge	Roles and authority in local government	A neighborhood association keeps hearing different officials referenced in discussions about a proposed project, and they are confused about who does what.	Clarify the roles of key local government bodies involved in land use decisions and explain how authority is divided among them.

Civic & Political Knowledge	Roles and authority in local government	A resident wants to get involved in a local issue but is unsure which government body to focus on.	Explain how different local boards and officials typically divide responsibilities, and how a resident can determine which one is most relevant.
Civic & Political Knowledge	Roles and authority in local government	A community group is preparing to engage with local government for the first time.	Provide a clear overview of how authority is structured across major local bodies and how their roles differ in decision-making processes.

Our full benchmark evaluation questions can be found here.

Appendix F: Additional Results





Figures above reflect 4o-mini judging results, grade ratio for each difficulty level and for each category respectively. More figures of other models are saved here.

Appendix G: Expert Evaluation Table

Overall Summary

***According to Peter**

#	Category	Subcategory	Score	%	What it did well*	What it didn't do well*
#001	Civic & Political Knowledge	Roles and authority in local government	14 / 20	70%	Correctly identified the main roles, explained the process sequence, and clearly defined terminology.	Too vague because responsibilities vary by jurisdiction. It needed regional-specific information, more civic leverage points, and more uncertainty about local variation.
#002	Civic & Political Knowledge	Roles and authority in local government	15 / 20	75%	Correctly identified major government bodies, explained the process sequence, and clearly defined terminology.	Generic answer. Focused on what government does, not how people can relate to it. It also should have acknowledged local variation more.
#021	Civic Information Access & Media Literacy	Finding official documents	15 / 20	75%	Evaluated source credibility, helped interpret civic documents, and avoided presenting unverified claims as facts.	Repeated document names already in the prompt. It needed to explain where a proposed policy change would be noted, how to gauge support, and how to track information better.
#041	Formal Participation in Government Processes	Speaking at meetings	20 / 20	100%	Strong across all categories: venue, rules, deadlines, action steps, evidence, follow-up, and accessibility.	No major weaknesses noted by Peter.
#081	Organizing & Associational Engagement	Finding and joining groups	12 / 20	60%	Included some useful steps for evaluating organizations, relationship-buildin	Less successful overall. Relied too much on online materials. It needed more talking to people, power analysis, group structure, internal conflict,

					g, and leadership development.	and decision-making guidance.
#101	Community Problem-Solving & Public Work	Volunteering opportunities	17 / 20	85%	Clearly defined the problem, identified assets, gave a feasible plan, identified partners, and included impact measurement.	Did not mention safety and equity. It also could have acknowledged that there may be no existing volunteer opportunities and suggested creating them.
#121	Relationships & Trust	Identifying relevant actors & networks	18 / 20	90%	Strong response. Identified relationships and gaps, emphasized reciprocity, included marginalized groups, and suggested ways to map networks.	Needed more specific relationship-building strategies, accountability, honesty, boundaries, and repair. It also may have offered too many suggestions, making it overwhelming.

Combined overall score: 111 / 140 = 79.3%

Peter's complete rubric tables can be found here.

Appendix H: Individual Reflections

Miriam Shamash

I am pleased to say that this class has been one of the best classes I have taken in graduate school. I will remember it as an example of what I hoped to gain from my MSCS degree: technical depth, meaningful application, and the chance to think critically about how computer science can be used in the world. I have always been interested in civic studies and social impact, so I was pleasantly surprised to see a course like this listed under the computer science offerings for the spring. I did not know exactly what to expect when I signed up, but a few months later, it has become one of the most meaningful classes I have taken at Tufts.

One of the things I appreciated most was how the class built both technical and conceptual literacy around AI and machine learning. The lectures, papers, quizzes, and project work helped bring everyone onto the same page while still challenging us to think deeply. I learned not only about generative AI, but

also about evaluation, responsible design, social impact, and the importance of grounding technical work in real human needs.

I am also exceptionally proud of the projects we built as a team. We started with Matthew, our tutor bot who, ironically, could not really do math, and ended with Sage, a civic chatbot that I could honestly see being deployed in the future with further testing and refinement. That growth made the class especially rewarding and showed me the potential of AI tools designed with care, context, and real social purpose. I also want to thank my teammates, Alisalee and Ning. We worked together not only on one project, but on two, and I am proud of everything we accomplished together this semester. Our projects improved because of the way we collaborated, challenged each other, divided the work, and kept pushing toward something stronger.

I also want to thank Fahad for the way he led the course. I was surprised by how meaningful each class session felt, and even more surprised by how well the discussions flowed without relying heavily on slides or a rigid structure. Somehow, a class about generative AI managed to feel both technically serious and very human, which is not always an easy balance. His feedback on our project was especially valuable, and I appreciated how he challenged us to go further in a way that felt constructive and motivating.

Thank you as well to the TAs for organizing the course, supporting the class, and spending time with us throughout the semester. Finally, I want to give an exceptional thank you to Peter, who guided us through the civic studies side of the project and generously donated his time and expertise. He was always kind, thoughtful, and genuinely excited to see our progress, which made each conversation with him both encouraging and useful. It was always a pleasure to speak with him, learn from his feedback, and see how his perspective helped us think more carefully about the civic purpose behind our work. Without his guidance, we would have been much more lost in the civic studies field, and his input made the project far more meaningful.

This class has made me very excited about the future of civic AI, and I am especially excited to see that future being explored at Tufts.

Alisalee Aaron

Prior to this class, I had a general knowledge about GenAI use but this class refined and expanded not only what I could do with GenAI but what I thought is and can be possible if GenAI applications are created carefully and with intention. One thing I valued was just how quickly we could put the theory we learned in class into practice. Our first project MATHew had us extensively use prompt engineering techniques and benchmarks to keep the chatbot focused and on topic.

Many of the lessons learned from that project directly influenced the prompting structure and system instructions of Sage. Many of the main features like generating citations and Ivy's web crawling are multi-shot procedures that utilize examples and structured output to increase response accuracy and make it easier to integrate with the hard coded systems around it. The papers we read in class also

directly inspired the creation of Orchid as an in-character persona-bot which filled the need for a dynamic yet consistent entity that can be used to simulate a person interacting with a CivicBot for FireBench. It is not often in class that every step of the way students can see where the readings and lecture material intersect in a way that they get to experience first hand in projects that speak to their interest instead of generic tasks which are completed by every student.

Having a professor who was passionate about his field and TA's who were always ready to support us made a real positive difference when it came to engaging with the content. Also, having the guidance of Peter on this project helped steer us in the correct direction in a domain that was previously unfamiliar to us, acting as both a mentor and a stakeholder in this project. Overall, I come away from this class with not only greater hope for the future of AI applications but the ability to shape it with the skills I've learned.

Ning Ni

In the last mini project, I was responsible for building a chatbot pipeline, so I chose to be responsible for the benchmark of this project for a different experience. I have used benchmarks such as Longbench on long-context understanding tasks and Advbench on safety alignment tasks, but do not know how they are built. As a result, I learned much more than what I expected to get. In the first version of benchmark, I mimic the design of Longbench that evaluates model performance by checking if LLM chooses the expected answer out of four candidates. However after meeting with the civic expert Prof. Peter, we realize a fixed correct answer is not suitable for an open-ended solution in the civic consulting area. I then discuss with my teammates for an alternative benchmark design and decide to use LLM-as-a-jury as we learn from lectures. It is a valuable experience to adapt system design based on feedback from our stakeholder, and strategically choose a benchmark approach following specific domain needs. This is the most professional AI agent project that I have participated in, and the best collaboration ever.